

MOHAMED MEHDI MECHKAK

APPLIED AI ENGINEER | PRODUCTION AUTOMATION SPECIALIST

Paris, France (CET) | +33 7 45 75 91 77 | mehdimechkak2025@gmail.com | [LinkedIn](#) | [GitHub Portfolio](#) | Bilingual: English (C2) & French (C2)

 **TARGET ROLE: AI ENGINEER / APPLIED AI DEVELOPER** | **AVAILABILITY: 20–30 HRS/WEEK (CONTRACT)** | **TIMEZONE: CET (EMEA & US OVERLAP)**

PROFESSIONAL SUMMARY

Applied AI Engineer and systems builder with a rigorous engineering foundation (Licence EEA at Université Paris-Saclay & Ecole Nationale Polytechnique). Specializing in production-grade AI systems, multi-lingual RAG architectures, FastAPI microservices, Airflow ETL data pipelines, and workflow automation. Available for 20–30 hour/week contract roles requiring CET timezone business hour coverage.

CORE TECHNICAL TOOLKIT (ATS KEYWORDS)

AI & LLM Engineering: RAG Architecture, Claude 3.5 API, OpenAI API, Vector DBs (pgvector, Pinecone), Prompt Engineering, Function Calling, LangChain, Structured Output (JSON Schema), Voiceflow.
Backend & Data Pipelines: Python 3.11, FastAPI, Apache Airflow (Python DAGs), Pandas, PostgreSQL, SQLAlchemy, REST APIs, Webhook Architecture, Streamlit, WeasyPrint PDF Engine.
Cloud & DevOps: Docker Compose, AWS (ECS Fargate, RDS, S3, CloudFront), Terraform (IaC), GitHub Actions CI/CD, Prometheus, Grafana, Linux/Bash.
Automation & Web: n8n, Make.com, Next.js 14, TypeScript, Playwright, Crawlee, Apify, Airtable, Google Sheets API.

FEATURED PRODUCTION AI & AUTOMATION PROJECTS

1. PolyLingual AI Support Agent Platform

2026
- FastAPI, PostgreSQL/pgvector, Claude 3.5, Next.js 14, Docker, AWS ECS Fargate, GitHub Actions CI/CD*
 - Engineered production multi-lingual (FR/EN) RAG support system with source citations, confidence gates, and human routing.
 - Deploys microservice backend using FastAPI and pgvector; containerized with Docker and infrastructure managed via Terraform on AWS.
2. Automated BI Pipeline with AI Enrichment

2026
- Apache Airflow, Python (Pandas/SQLAlchemy), PostgreSQL, Streamlit, WeasyPrint, Docker*
 - Architected Airflow-orchestrated ETL pipeline parsing 5 data sources into PostgreSQL marts with AI classification, review summarization, and anomaly detection.
 - Delivered bilingual Streamlit executive dashboard and automated PDF report generation with Slack alerts.
3. AI-Powered Internal Tool Builder

2026
- Next.js, TypeScript, Claude API (Structured JSON Schema), AWS Lambda, n8n Webhooks*
 - Built a natural language automation builder converting non-technical prompt specifications into executable n8n & AWS Lambda workflows.
4. ConciergeFlow AI – Hotel Automation Platform

2024 – 2026
- n8n, Voiceflow, Streamlit, Playwright, Make.com, REST APIs*
 - Created interactive multi-lingual AI concierge for Paris boutique hotels handling recommendations, quick replies, and staff dispatching.
5. Lead Scraper & AI Cold Outreach Engine

2024 – 2026
- Python, Crawlee, Playwright, Apify, Groq API / OpenAI, Airtable API*
 - Built automated lead extraction engine with AI personalization drafting customized outreach copy into Airtable tracking databases.
6. Market Intelligence Data Scraper & ETL Engine

2024 – 2026
- Python, Pandas, Streamlit, Web Scraping, REST APIs*
 - Developed real-time market data extraction and transformation dashboard for competitive intelligence monitoring.

INDUSTRIAL ENGINEERING EXPERIENCE

- Engineering Internships – Power Systems & Industrial Automation

2023 – 2025
- SONATRACH / Equinor / ENI / CEPSA / Al-Hikmah Pharmaceuticals*
 - High-reliability industrial automation environments: DCS/PLC/HMI supervision, electrical protection systems, relay testing, and maintenance controls.

TEACHING & EDUCATION

- Licence 3, Electronics, Electrical Energy & Automation (EEA)

2025 – Present
- Université Paris-Saclay, Orsay, France | Focus: Computer Science, Automation & Networks*
- Engineering Cycle, Electrical Engineering & Hydrogen Tech

2021 – 2025
- Ecole Nationale Polytechnique, Algiers*